

Tahsin Raihan Robbani

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Education

BRAC University

BSc in Computer Science and Engineering—CGPA: 3.83/4.00

Dhaka, Bangladesh
Sep 2022 – Sep 2026 (Expected)

Notre Dame College

Higher Secondary Certificate (HSC) — GPA: 5.00/5.00

Dhaka, Bangladesh
2019 – 2021

National Ideal School

Secondary School Certificate (SSC) — GPA: 5.00/5.00

Dhaka, Bangladesh
2019

Experience

Student Tutor (ST)

BRAC University – Department of Computer Science and Engineering

Mar 2026 – Present
Dhaka, Bangladesh

- Serve as a Student Tutor across **Spring 2026** and **Summer 2026**, assigned to **CSE230 (Discrete Mathematics)** course.
- Hold regular consultation hours to help students work through difficult proofs, set theory, graph theory, and complex mathematical logic.
- Guide students step-by-step through complex practice problems and theoretical concepts during 1-on-1 and group consultations.
- Coordinate regularly with course faculties to align consultation topics with lecture progress and address common student difficulties.

Technical Skills

Programming Languages: Python, Java, JavaScript, C, SQL, HTML.

Backend & Databases: Node.js, PostgreSQL, MongoDB, Supabase, Vercel, Git.

Frontend: Svelte, React, HTML/CSS.

AI & Computer Vision: Deep Learning, Image & Video Processing, Bioinformatics, SimCLR.

Machine Learning: YOLOv8, RT-DETR, U-Net++, MedSAM2, MediaPipe, I3D.

Graphics, Hardware & Tools: OpenGL, Arduino, LaTeX.

Research

Neuro-Symbolic AI for Optical Biopsy-Level Polyp Diagnosis: Autonomous Classification, Risk Prediction, and Interpretable Reasoning in Endoscopic Video

2025 – 2026

Undergraduate Thesis | Python, YOLOv8, RT-DETR, MedSAM2, SimCLR

- Built a multi-stage Neuro-Symbolic AI framework combining real-time detection models (YOLOv8, RT-DETR) and segmentors (MedSAM2) to analyze endoscopic video for colorectal cancer screening.
- Achieved over **93% classification accuracy** by incorporating consensus-based detection and calibrated probability models to deliver interpretable, real-time diagnostic risk assessments.

Engineering Projects

Oi Mama (Banglish DSL Compiler) | *Python, AST, Lexer/Parser, PyPI, CLI* 2026

- Built and published a custom Banglish Domain-Specific Language (DSL) live on PyPI, enabling code execution using everyday Bangladeshi slang.
- Designed a smart Lexer and leveraged Python's AST engine to support OOP, async execution, and custom Banglish error reporting.

Gesture Board (Real-Time App) | *React, Supabase, MediaPipe, Vercel* 2026

- Engineered a low-latency multi-user whiteboard deployed on Vercel with touchless gesture-based drawing and navigation.
- Implemented continuous spatial coordinate mapping using MediaPipe hand tracking for real-time interaction.

Multi-Cancer Classification | *Python, PyTorch, Torchvision, CNN* 2026

- Built a deep learning pipeline using CNNs and transfer learning to classify cancer typologies from genomic sequence datasets.
- Tuned hyperparameters and data augmentation strategies to optimize multiclass classification accuracy and model generalization.

UniEats (Campus Food Platform) | *Svelte, TypeScript, PLpgSQL* 2024

- Developed a campus-focused food ordering platform featuring responsive Svelte frontend logic and optimized database querying.

Zombie Survival Game | *Python, OpenGL* 2023

- Programmed an interactive survival game featuring custom 2D/3D collision detection, dynamic enemy pathfinding AI, and resource management.

Universe (Interactive Web Application) | *Svelte, TypeScript, PLpgSQL* 2024

- Engineered an interactive application focusing on scalable system architecture, seamless UI/UX, and database efficiency.

File System Metadata Journaling | *C, Systems Programming, POSIX* 2024

- Engineered a write-ahead metadata journaling system for a virtual file system (VSFS) to ensure crash consistency.
- Implemented atomic transaction logging and automated recovery mechanisms to prevent inode and bitmap corruption upon failure.

Extra-Curricular Activities

Panel Member & Volunteer 2020 – 2021
Notre Dame Information and Technology Club (NDITC) Dhaka, Bangladesh

- Managed event logistics, schedules, and technical infrastructure for "iNit 2.0," a national technology festival.
- Handled inter-school communications and crowd management as part of the core organizing panel.